

Doyelshree Bhui

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Education

Techno Main Salt Lake <i>Bachelor of Technology, Information Technology</i>	<i>Nov 2022 - Present</i>
B.D.M.I Kolkata <i>Higher Secondary</i>	<i>Apr 2009 - Jun 2022</i>

Experience

Research Intern <i>IEEE Computational Intelligence Society</i> 🔗	<i>Kolkata, WB</i> <i>June 2025 - July 2025</i>
- Studied evolutionary computation techniques for optimizing ligand structures using Genetic Algorithms (GAs) in drug design - Minimized average interaction energy between ligand molecules and protein active sites using a variable-length tree model - Developed and published a complete implementation of the NBGA algorithm using TSPLIB datasets to generate benchmark graphs and validate results (Reference Paper 🔗, Source Code 🔗)	

Projects

Smart India Hackathon 2024	Source Code 🔗
Achieved 2nd place among 500+ teams with 5 peers in Punjab, as a full stack developer developed a mentorship platform serving 1000+ potential users - Engineered mentor-mentee matching algorithms achieving 92% compatibility accuracy, real-time scheduling handling 50+ concurrent sessions, and AI-driven career guidance - Tech Stack: React.js, Tailwind CSS, Flask, Python, Docker, Firebase, Cal.com API, OpenAI API	
Cardiovascular Disease Risk Prediction - IoT + ML	Source Code 🔗
- Built an end-to-end system predicting cardiovascular disease risk by combining live IoT sensor readings (ECG and pulse oximetry via ESP32) with manually entered patient details - Developed a React Native mobile app for data collection and a Flask backend to aggregate USB serial sensor samples and serve real-time predictions - Trained an SVM classification model using scikit-learn on a Heart Failure dataset with 11 features, implementing population median imputation for missing clinical data	
AWS S3 File Uploader	Source Code 🔗
- Built a full-stack web application with Spring Boot (Java) and React for seamless file management - Integrated AWS S3 cloud storage for secure file uploads along with authentication, validation, and responsive UI - Tech Stack: Java, Spring Boot, Maven, React, TypeScript, AWS S3	
Farmora (Agricultural Disease Detection Platform)	Source Code 🔗
- Trained on a dataset of 10,000+ labeled plant images, achieving 88% classification accuracy across 38 disease classes empowering farmers to reduce crop losses and improve yield quality - Developed a full-stack web application with ML backend and Gemini AI chatbot for real-time crop advice - Tech Stack: Python, JavaScript, HTML/CSS, REST API	

Technologies

Languages: C, Java, Python, SQL, JavaScript, TypeScript, HTML, CSS
Frameworks/Tech: React Native, Expo, Flask, scikit-learn, Spring Boot, React.js, AWS S3, IoT (ESP32), Machine Learning (SVM), Git, Linux